Fertility and Sterility

# Concordant mtDNA heteroplasmy between trophectoderm and inner cell mass validates blastocyst-stage PGT-mt for m.14487T>C

--Manuscript Draft--

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| **Manuscript Number:** | |
| **Article Type:** | Laboratory-based study |
| **Keywords:** | PGT-mt, mitochondrial DNA, T 14487 C, MT-ND6, heteroplasmy, Leigh syndrome, ICM, TE |
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| **Abstract:** | Objective: Is blastocyst-stage trophectoderm (TE) biopsy informative for preimplantation genetic testing of the mitochondrial DNA (mtDNA) variant m.14487T>C (MT-ND6), by providing heteroplasmy estimates representative of the inner cell mass (ICM), within a combined PGT-A/PGT-mt workflow?<br>Design: Single IVF/ICSI cycle study in one carrier woman; six blastocysts were obtained and underwent Day 5–6 TE biopsy followed by combined PGT-A and targeted mtDNA heteroplasmy assessment.<br>Setting: Fertility center. Diagnostics laboratory.<br>Patient: A 38-year-old woman carrying the pathogenic mtDNA variant m.14487T>C underwent IVF/ICSI and blastocyst culture. TE biopsies were processed for PGT-A using NGS after whole-genome amplification, and m.14487T>C heteroplasmy was quantified by deep sequencing of an amplicon spanning nucleotide 14487. For a subset of embryos donated to research, isolated TE and ICM fractions were analyzed separately.<br>Interventions: Embryo biopsy, genetic testing, inner call mass, concordance<br>Main Outcome Measures: Test accuracy, treatment outcomes, and mutation segregation.<br>Results: Six blastocysts underwent combined PGT-A/PGT-mt (Day 5, n=2; Day 6, n=4). PGT-A classified three embryos as euploid, two as aneuploid and one as mosaic. PGT-mt showed a bimodal distribution of m.14487T>C heteroplasmy: three embryos were <18% (11.4–13.5%) and three were >70% (73.2–99.6%); only one embryo met the predefined transfer criteria (euploid; 11.4% heteroplasmy). After warming and embryo fractionation, heteroplasmy estimates from the original TE biopsy and the corresponding post-warming embryo fraction (the remaining TE, ICM, or combined TE+ICM) did not differ significantly.<br>Conclusion: These findings provide variant-specific evidence that blastocyst TE biopsy can be representative of the ICM for m.14487T>C, supporting combined PGT-A/PGT-mt at the blastocyst stage. Nevertheless, given the possibility of heteroplasmy shifts later in development, prenatal diagnosis and postnatal follow-up remain advisable |

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| | when a heteroplasmic embryo is transferred. |
| Opposed Reviewers: | |

Dear Editors,

We submit our manuscript for consideration in Fertility and Sterility as a Research Article entitled: “Concordant mtDNA heteroplasmy between trophectoderm and inner cell mass validates blastocyst-stage PGT-mt for m.14487T>C”.

Preimplantation genetic testing for mtDNA disorders requires quantitative assessment of mutation load, and the representativeness of a small embryo biopsy remains a key practical concern. In a clinically grounded dataset from a carrier woman with high tissue heteroplasmy and a family history consistent with Leigh syndrome, we performed combined PGT-A and targeted quantification of m.14487T>C at the blastocyst stage in six embryos. The couple decided to donate all embryos to research: they consented to additional molecular analyses to evaluate the distribution of m.14487T>C heteroplasmy between ICM and the remaining TE. We fractionated the embryos and re-measured heteroplasmy in the fractions (ICM or combined TE+ICM where separation was not feasible, and remaining TE). Paired analyses showed no significant differences between TE biopsy and the fractions with high concordance. These findings provide variant-specific evidence that blastocyst TE biopsy can be representative of the ICM for m.14487T>C, supporting combined PGT-A/PGT-mt at the blastocyst stage.

The manuscript is original, not under consideration elsewhere, and approved by all authors. Ethics approval and written informed consent are described in the manuscript.

Thank you for your consideration.

Sincerely,

Ilya Mazunin, PhD
on behalf of all authors

Manuscript Click here to view linked References

**Running Title:** Blastocyst-stage PGT-mt is enabled by concordant mtDNA heteroplasmy

**Article Title:** Concordant mtDNA heteroplasmy between trophectoderm and inner cell mass validates blastocyst-stage PGT-mt for m.14487T>C

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**Article Type:** Laboratory-based study

**Funding Statement:** N/A

**Disclosure Statement:** The authors have no disclosures or conflicts of interest to declare.

**Attestation Statement:** Data regarding any of the subjects in the study has not been previously published. Available data will be made available to the editors of the journal for review or query upon request.

**Data Sharing Statement:** All scripts and all data corresponding to mtDNA variant analysis are openly available on Github (https://eugot.github.io/m14487/heteroplasmy_stability.html)

**Word Count:** ЧЧ for abstract and ЯЯ for text

**Capsule:** Pathogenic m.14487T>C variant showed close concordance between TE and ICM in embryos with paired analyses, supporting the feasibility of blastocyst-stage PGT-mt integrated with PGT-A for this Leigh syndrome-associated variant.

**Structured Abstract**

**Objective:** Is blastocyst-stage trophectoderm (TE) biopsy informative for preimplantation genetic testing of the mitochondrial DNA (mtDNA) variant m.14487T>C (*MT-ND6*), by providing heteroplasmy estimates representative of the inner cell mass (ICM), within a combined PGT-A/PGT-mt workflow?

**Design:** Single IVF/ICSI cycle study in one carrier woman; six blastocysts were obtained and underwent Day 5–6 TE biopsy followed by combined PGT-A and targeted mtDNA heteroplasmy assessment.

**Setting:** Fertility centers. Diagnostics laboratory.

**Patient:** A 38-year-old woman carrying the pathogenic mtDNA variant m.14487T>C underwent IVF/ICSI and blastocyst culture. TE biopsies were processed for PGT-A using NGS after whole-genome amplification, and m.14487T>C heteroplasmy was quantified by deep sequencing of an amplicon spanning nucleotide 14487. For a subset of embryos donated to research, isolated TE and ICM fractions were analyzed separately.

**Interventions:** Embryo biopsy, genetic testing, inner call mass, concordance

**Main Outcome Measures:** Test accuracy, treatment outcomes, and mutation segregation.

**Results:** Six blastocysts underwent combined PGT-A/PGT-mt (Day 5, n=2; Day 6, n=4). PGT-A classified three embryos as euploid, two as aneuploid and one as mosaic. PGT-mt showed a bimodal distribution of m.14487T>C heteroplasmy: three embryos were <18% (11.4–13.5%) and three were >70% (73.2–99.6%); only one embryo met the predefined transfer criteria (euploid; 11.4% heteroplasmy). After warming and embryo fractionation, heteroplasmy estimates from the original TE biopsy and the corresponding post-warming embryo fraction (the remaining TE, ICM, or combined TE+ICM) did not differ significantly.

**Conclusion:** These findings provide variant-specific evidence that blastocyst TE biopsy can be representative of the ICM for m.14487T>C, supporting combined PGT-A/PGT-mt at the blastocyst stage. Nevertheless, given the possibility of heteroplasmy shifts later in development, prenatal diagnosis and postnatal follow-up remain advisable when a heteroplasmic embryo is transferred.

**Key Words:** PGT-mt, mitochondrial DNA, T 14487 C, MT-ND6, heteroplasmy, Leigh syndrome, ICM, TE

## INTRODUCTION

Mitochondrial DNA (mtDNA) is a circular, double-stranded genome located in mitochondria and transmitted almost exclusively through the maternal lineage. Its copy number varies markedly across cell types, being extremely low in sperm and reaching hundreds of thousands of copies in mature oocytes (Liu *et al.,* 2025). Because mtDNA encodes essential subunits of the oxidative phosphorylation system, pathogenic mtDNA variants most commonly present with energy-demanding tissue involvement, including neuromuscular and neurodegenerative phenotypes (Suomalainen and Nunnari, 2024).

Leigh syndrome is one of the most severe clinical manifestations of mitochondrial disease and typically leads to early-onset, progressive neurodegeneration with high childhood mortality (Rahman, 2023). As current management is largely supportive, preventing transmission of pathogenic mtDNA variants remains a major goal of reproductive care (Castelluccio *et al.,* 2025). In contrast to Mendelian inheritance of nuclear variants, mtDNA disorders follow distinct rules of mitochondrial genetics that complicate risk prediction and embryo selection. During oogenesis, a mitochondrial "genetic bottleneck" results in substantial variation in heteroplasmy levels among individual oocytes from the same woman, and the extent of this variation may depend on the specific variant. Disease expression is primarily driven by heteroplasmy, defined as the coexistence of mutant and wild-type mtDNA, whereas homoplasmy denotes a uniform mtDNA sequence. Clinical and biochemical dysfunction typically occurs only once heteroplasmy exceeds a variant- and tissue-dependent threshold. In addition, heteroplasmy may differ between tissues due to mitotic segregation, contributing to marked phenotypic variability (van den Ameele *et al.*, 2019). Consequently, even low heteroplasmy in an asymptomatic carrier does not exclude the presence of oocytes with a high mutational load, and embryo selection criteria must be tailored to the causative variant and the available clinical evidence. Unlike preimplantation genetic testing for monogenic nuclear disorders (PGT-M), PGT for mtDNA variants (PGT-mt) requires not only variant detection but also quantitative assessment of heteroplasmy in the embryo biopsy. A conservative heteroplasmic cutoff around 18% is often cited in counseling, although thresholds are not universal and may differ across variants and tissues (Hellebrekers *et al.,* 2012).

Cleavage-stage biopsy using one or two blastomeres (day 3) remains a commonly used approach for PGT-mt, as reflected by a 15-year single-centre experience in France (cycles 2006–2021) (Mayeur *et al.,* 2023). Blastocyst-stage trophectoderm (TE) biopsy has also been implemented for PGT-mt, particularly within combined workflows integrating pathogenic mtDNA variants detection with PGT-A to prioritize euploid embryos with low heteroplasmy (Späth *et al.,* 2021).

Here, we report heteroplasmy measurements for the pathogenic mtDNA variant m.14487T>C in *MT-ND6*, associated with Leigh syndrome, obtained from TE biopsy at the blastocyst stage in a protocol combined with PGT-A. Importantly, we further assessed the distribution of m.14487T>C between TE and the inner cell mass (ICM) in the same embryos, including both euploid and aneuploid blastocysts. The variant m.14487T>C has been reported in a limited number of families, with heteroplasmy levels varying across tissues, ages, and clinical presentations, but generally exceeding commonly used counseling thresholds. To our knowledge, this is the second reported PGT-mt case for m.14487T>C and the first to evaluate this variant at the blastocyst stage with paired TE and ICM analysis. Given the limited number of clinical observations with postnatal validation after PGT-mt, additional well-characterised reports remain essential to refine the reliability of the approach and to inform embryo selection strategies.

## MATERIALS AND METHODS

### Ethics

The study was approved by the Ethics Committee of Fomin Clinic, Moscow, Russia (ID No: 03/2021). The couple agreed to donate all their embryos to research and provided written informed consent.

### Medical history

A 37-year-old female was found to carry the pathogenic mtDNA variant m.14487T>C in *MT-ND6* with heteroplasmy of approximately 38% in peripheral blood lymphocytes, 86% in urine, and 90% in skeletal muscle. She had no overt neurological manifestations, reporting only intermittent muscle weakness. Her first child (female, proband), conceived spontaneously and delivered at term in 2010, was initially considered healthy. Hypotonia was noted at 2 months of age. By 18 months, psychomotor delay and ataxia became apparent. At 2 years, she developed developmental regression with loss of independent ambulation and progression to spastic tetraparesis. Laboratory testing showed elevated lactate and pyruvate levels, and brain MRI demonstrated a neurodegenerative pattern consistent with Leigh syndrome. The variant m.14487T>C was detected in the child's lymphocytes at approximately 50% heteroplasmy. Despite intensive supportive care, she developed bulbar dysfunction with feeding difficulties and, from 4 years of age, epileptic seizures. She died at the age of 6 following an acute metabolic decompensation triggered by a respiratory infection. The female's second child (female), conceived spontaneously and born in 2013, carried the same variant with heteroplasmy of approximately 35% in lymphocytes and 41% in urine; as of February 2026, she remained clinically asymptomatic with no features of Leigh syndrome. In 2017, the female experienced a third spontaneous pregnancy with a male fetus, which ended in miscarriage at 20 weeks' gestation. Subsequently, she had a healthy daughter born in 2020 following IVF with donor oocytes. The woman and her 38-year-old husband sought IVF with PGT-mt to conceive a genetically related child while minimizing the risk of mtDNA disease transmission (Figure 1).

The couple received genetic counseling regarding a combined workflow of PGT-A/PGT-mt performed on TE biopsies at the blastocyst stage. The patients were also informed that, to date, only one PGT-mt case for the pathogenic variant m.14487T>C has been reported, and this procedure was performed at the cleavage stage (Sallevelt *et al.,* 2017). Written informed consent was obtained; the consent specified an 18% heteroplasmy threshold for embryo transfer. At the time the IVF cycle was initiated, the woman was 38 years old and had no medical contraindications to ovarian stimulation or IVF.

### Ovarian stimulation

Ovarian stimulation was performed with the progestin-primed ovarian stimulation protocol, started on the 2nd day of the spontaneous menstrual cycle. The patient received one single subcutaneous injection of corifollitropin alpha 150 mcg (Elonva) and on the same day, she started taking dydrogesterone 10 mg twice a day till the day of trigger injection. On the $8^{th}$ and $9^{th}$ days of stimulation, she received menotropins (Humog) 150IU once a day subcutaneously. Agonist trigger triptorelin 0.2 mg (Decapeptyl) was administered 37 hours before the ovum pickup. Totally 19 cumulus-oocyte complexes were obtained from 21 follicles.

### Embryo culture and TE biopsy (for PGT-A/PGT-mt)

ICSI was performed on MII oocytes (n=12). Fertilized oocytes (n=6) were cultured up to the blastocyst stage (day 5–6) and graded using the Gardner–Schoolcraft system. TE biopsy was performed for two blastocysts on day 5 and for four blastocysts on day 6; biopsied cells were used for combined PGT-A/PGT-mt, and all blastocysts were cryopreserved.

For combined PGT-A/PGT-mt, laser-assisted trophectoderm (TE) biopsy was performed on each viable blastocyst, removing ~5–10 TE cells. Biopsy was carried out on an inverted microscope (Olympus IX71) equipped with Narishige micromanipulators, using a holding pipette (Cook K-HPIP-3335; Cook, Vandergrift, USA) and biopsy pipette (Cook K-EBPH-3035; Cook, Vandergrift, USA). Embryos were handled in drops of Handling with phenol red (Origio, CooperSurgical, Denmark) under Oil for tissue culture (Origio, CooperSurgical, Denmark). The intended

incision site was gently sucked into the biopsy pipette, marked with a laser (Octax; 2-µs pulse), and the TE fragment was detached by gentle mechanical flicking. Following biopsy, blastocysts were vitrified using the Kitazato media.

**PGT-A/PGT-mt**

For PGT-A, sequencing libraries were prepared following whole-genome amplification (WGA) using the AB Vector AB-PGT workflow and sequenced on an Illumina MiSeq platform. FASTQ files were aligned to GRCh38, and copy-number analysis was performed using CNVector tools (AB Vector) (Belyaev *et al.,* 2024). Clinical reporting was based on CNVector chart-based interpretation (Supplementary Figure 1). Embryos were classified as recommended or not recommended for transfer according to their chromosomal status. Euploid embryos and embryos with low-level mosaicism (<20%) were recommended for transfer. Embryos with aneuploidy, deletions/duplications, or high-level mosaicism (>80%) were not recommended for transfer. Embryos with mosaic aneuploidy involving chromosomes 13, 14, 16, 18, 21, or 45,X were not recommended regardless of the estimated mosaicism level. For embryos with a single mosaic aneuploidy between 20% and 80%, genetic counseling was recommended to assess transfer-related risks.

PCR primers were designed to amplify an mtDNA fragment spanning the pathogenic variant m.14487T>C. Targeted amplification was performed to achieve high read depth at the variant site. The resulting amplicons were incorporated into the sequencing library (together with the WGA-derived material) and sequenced. Heteroplasmy was quantified as the variant allele fraction, calculated as the proportion of reads carrying the alternate allele among all high-quality reads covering position 14487 (Supplementary Code). Primer sequences are provided in Supplementary Table 1.

**Postbiopsy fate of the embryos**

After obtaining the PGT-A/PGT-mt results, the patients underwent medical consultations and genetic counselling. The couple elected not to proceed with embryo transfer due to concerns regarding potential clinical risks associated with uneven distribution of mtDNA pathogenic variants. Thus, the couple decided to donate all 6 obtained embryos to research. Specifically, they consented to additional molecular analyses to evaluate the distribution of m.14487T>C heteroplasmy between ICM and the remaining TE in their embryos.

Vitrified blastocysts were warmed using the Kitazato Ultra-Fast Warming kit (VT602UF; Kitazato, Japan) into a pre-equilibrated KSOM-based single-step medium (Ural-m-1step; Inkamedik, Russia). Because post-warming recovery varied between embryos, all embryos were cultured for an additional 21 h prior to sampling; images were acquired immediately after warming and at 1.5 h and 21 h post-warming.

After recovery culture, blastocysts were removed from the zona pellucida (ZP) and, where feasible, mechanically separated into two fractions: ICM+ (the inner cell mass together with adherent/polar TE inseparable from the ICM) and the remaining TE fraction. If a clean separation was not feasible due to the embryo's low quality, the whole embryo was processed as a single sample. The ZP with any excluded cells/cellular fragments (EC) retained under or attached to it was collected separately (Supplementary Figure 2; Supplementary Videos). Dissection was performed under an inverted microscope (Nikon Eclipse Ti) equipped with Narishige micromanipulators, using a holding pipette (Cook K-HPIP-3335; Cook, Vandergrift, USA) and biopsy pipette (Cook K-EBPH-3035; Cook, Vandergrift, USA). The ICM+TE fraction was aspirated into the biopsy pipette and detached from the remaining TE by controlled mechanical flicking after laser marking of the incision site.

**Statistical analysis**

Agreement analysis — Lin's concordance correlation coefficient (CCC) and Bland–Altman limits of agreement for two explicit primary pairings: PGT biopsy vs ICM and PGT biopsy vs remaining TE, with EC analyzed separately as exploratory. Repeated-measures design — Friedman's test (non-parametric alternative to repeated-measures ANOVA) for within-embryo fraction effects, with Kendall's W as effect size. Estimation statistics — Gardner–Altman paired mean-difference plots with bootstrap 95% confidence intervals, following the DABESTR philosophy of prioritising effect sizes over p-values. Detailed statistics are available on github (https://eugot.github.io/m14487/heteroplasmy_stability.html).

## RESULTS

During ovarian stimulation, 19 oocytes were retrieved: 12 MII, 3 MI, 3 GV, and 1 degenerate oocyte. All MII oocytes underwent ICSI. Fertilization assessment identified six 2PN zygotes and six 0PN oocytes/zygotes; four of the 2PN embryos and two of the 0PN embryos developed to the blastocyst stage.

Chromosomal status and m.14487T>C heteroplasmy results for each embryo are summarized in Table 1. PGT-A classified three embryos as euploid, two as aneuploid, and one as mosaic. PGT-mt revealed a bimodal distribution of heteroplasmy: three embryos showed low mutant loads (<14%), whereas three embryos showed high mutant loads (>73%). In this small dataset, the heteroplasmy distribution did not show an obvious association with embryo sex or with chromosomal status/specific aneuploidies detected by PGT-A; however, the sample size is too limited to support formal inference. When integrating PGT-A and PGT-mt results, only one embryo met the predefined transfer criteria (euploid with heteroplasmy level of 11.4%) and was therefore recommended for transfer.

After receiving the combined PGT-A/PGT-mt results, the couple elected not to proceed with the transfer. Although the recommended embryo met the predefined heteroplasmy threshold, the couple expressed concern regarding the residual presence of the pathogenic variant and decided not to use it. To further evaluate the representativeness of TE-based heteroplasmy estimates (and to inform genetic family counseling), all six embryos were donated for additional research and molecular analyses.

Paired heteroplasmy estimates for m.14487T>C from the original TE biopsy were evaluated in two explicit primary comparisons: TE biopsy versus ICM, and TE biopsy versus remaining TE (n = 4 embryos with both fractions available: embryos 1, 2, 5, and 6). For TE biopsy versus ICM, concordance was very high (Pearson r = 0.9996; Lin's CCC = 0.9994; Cb = 0.9999; Figure 2, top row; Figure 5). The mean paired difference (TE biopsy minus ICM) was +0.707 percentage points (pp), with Bland-Altman 95% limits of agreement from -1.90 to +3.31 pp (Figure 2, top right), exact two-sided signed-rank p = 0.3750, and BCa 95% confidence interval for the mean difference from -0.590 to +1.775 pp (Figure 3).

For TE biopsy versus remaining TE, concordance was also high (Pearson r = 0.9979; Lin's CCC = 0.9970; Cb = 0.9991; Figure 2, bottom row; Figure 5). The mean paired difference (TE biopsy minus remaining TE) was -1.410 pp, with Bland-Altman 95% limits of agreement from -7.69 to +4.87 pp (Figure 2, bottom right), exact two-sided signed-rank p = 0.6250, and BCa 95% confidence interval for the mean difference from -4.716 to +0.467 pp (Figure 3).

Because EC can include excluded cellular material and mixed debris under/attached to the zona pellucida, EC analyses were treated as exploratory. TE versus EC showed a median absolute difference of 2.929 pp (exact signed-rank p = 0.1562). EC versus ICM showed a median absolute difference of 4.704 pp (exact signed-rank p = 0.2500), and EC versus remaining TE showed a median absolute difference of 5.243 pp (exact signed-rank p = 0.6250). These exploratory comparisons are visualised in the full pairwise concordance matrix (Figure 5).

In the balanced repeated-measures subset (ICM, TE biopsy, remaining TE, EC; n = 4), Friedman testing yielded chi-square(3) = 4.500 with p = 0.2123 by chi-square approximation and p = 0.2417 by exact permutation, with Kendall's W = 0.375 (moderate effect size). Within-embryo heteroplasmy profiles were remarkably flat across fractions (Figure 4A), with deviations from the embryo mean generally within ±5 pp (Figure 4B). Pairwise Gardner-Altman estimates (BCa 95% CIs) indicated that some CIs excluded zero, but exact signed-rank p-values for all pairwise contrasts remained >= 0.125 in this small sample (Figure 3).

A Bayesian blocked repeated-measures model (embryo effects + fraction effects, weakly informative priors) provided directionally consistent estimates while explicitly representing uncertainty for fraction contrasts. Depth-aware posterior sensitivity for primary pairings remained tightly centered near small effects for TE vs ICM (95% interval -0.770 to +2.118 pp; P(|mean diff| <= 2 pp) = 0.964; P(|mean diff| <= 5 pp) = 1.000), while TE versus remaining TE showed broader uncertainty and a negative-centered posterior mean difference (95% interval -2.518 to -0.146 pp).

Overall, the primary analyses directly support high concordance between TE-biopsy heteroplasmy and both ICM and remaining-TE fractions in this dataset, with the strongest precision for the TE-versus-ICM pairing (Figure 2; Figure 5). However, precision remains limited by sample size, and EC findings remain exploratory.

## DISCUSSION

The key finding of this study is the high concordance of m.14487T>C heteroplasmy estimates between the original TE biopsy and the other embryonic fraction (TE, ICM, or combined TE+ICM where separation was not feasible). While the small sample size limits statistical power, the absence of a consistent directional shift supports the practical use of blastocyst-stage TE biopsy for heteroplasmy quantification of m.14487T>C within a combined PGT-A/PGT-mt workflow.

Our data also suggest that even in a woman with high heteroplasmy for m.14487T>C across tissues (~38% in lymphocytes, ~86% in urine, and ~90% in muscle), it may still be possible to obtain embryos with heteroplasmy below the commonly used transfer threshold of 18%. In this setting, maternal age is a critical determinant of feasibility, as the probability of obtaining euploid embryos declines substantially after age 35. Consistent with this, Volodyaev et al. estimated that whole-chromosome aneuploidy increases by ~6% per year after age 35 (with an euploidy rate of ~35% at age 35), which directly limits the number of embryos available for simultaneous selection by chromosomal status and mtDNA heteroplasmy (Volodyaev *et al.*, 2025).

An additional observation in our dataset was the bimodal distribution of embryo heteroplasmy for m.14487T>C, with embryos clustering at low and high mutational loads. This pattern contrasts with the previously published PGT-mt report for m.14487T>C, in which most embryos exhibited high heteroplasmy (Sallevelt *et al.,* 2017). Variant-specific heteroplasmy distributions have been described across mtDNA disorders, reflecting the mitochondrial genetic bottleneck and the fact that different mutations can show distinct segregation patterns during oogenesis (Chinnery *et al.,* 2000). With larger patient datasets, it has become apparent that some variants yield a broad continuum of heteroplasmy values among oocytes, whereas others tend toward bimodality, which has direct implications for the likelihood of identifying embryos below a given cutoff.

Importantly, the central limitation for the clinical interpretation of PGT-mt is not merely the ability to identify embryos with low heteroplasmy, but whether a small biopsy sample is truly representative of the remaining embryo and, ultimately, of the offspring in terms of heteroplasmy. If the biopsy sample is shown not to be representative, alternative approaches such as mitochondrial replacement therapy (MRT) may be considered to prevent the transmission of mtDNA disease (Hyslop et al., 2025). However, MRT for mitochondrial diseases also poses significant technical, legal, and ethical challenges and may therefore not be the first option of choice for the majority of patients with mitochondrial disease (Kirillova and Mazunin, 2022). This concern applies to both cleavage-stage PGT-mt based on one or two blastomeres and blastocyst-stage PGT-mt based on 5–10 TE cells. At the cleavage stage, overall concordance of heteroplasmy between blastomeres is generally high; for m.14487T>C specifically, Sallevelt et al. reported differences of <10% between blastomeres in most embryos (17/19), with larger discordance (20–30%) observed in a minority (Sallevelt *et al.*, 2017). Similar patterns have been reported for other pathogenic mtDNA variants (see reviews, Castelluccio *et al.,* 2025; Lan *et al.,* 2024).

At the blastocyst stage, accumulating evidence supports the use of TE biopsy for estimating mtDNA heteroplasmy in an embryo. Spath et al. developed an NGS-based approach that quantifies patient-specific mtDNA mutations directly from the same WGA product generated for routine PGT-A, enabling integrated PGT-A/PGT-mt workflows (Spath *et al.,* 2021). Janssen et al. introduced a clinical-grade WGS-PGT framework and reported that heteroplasmy estimates from day-5 TE biopsies differed by only 1–4% from day-3 PCR-RFLP measurements, with similarly small differences for surplus embryo material; WGS-PGT produced closely matching values and remained robust under subsampling (Janssen *et al.,* 2024). In addition, Rubens et al. assessed mtDNA heteroplasmy across preimplantation stages and within blastocyst compartments and observed high concordance between TE and ICM (Rubens *et al.,* 2025). Our paired TE-ICM measurements for m.14487T>C are consistent with these observations. Mechanistically, deep-sequencing studies of mtDNA across oocytes, blastomeres, and blastocyst fractions suggest that variants recurring across multiple samples from the same embryo typically show highly similar heteroplasmy loads, supporting largely homogeneous partitioning during early human development (Mertens *et al.,* 2022).

The EC fraction is expected to be heterogeneous and may include intact extra cells, cellular fragments without nuclei, and degenerating debris. Consistent with this, PGT-A profiles derived from such samples were noisier than those from TE/ICM fractions, and some EC samples showed patterns compatible with DNA degradation (Supplementary Table 1). In line with the CNV noise, heteroplasmy estimates in the EC fraction showed higher variability compared with the TE-ICM comparisons. The greater dispersion of the EC fraction is visually apparent in the within-embryo profiles. Therefore, we interpret EC measurements as exploratory and do not consider this compartment suitable for clinical decision-making, in contrast to TE biopsy and embryo-derived fractions.

Discordant findings have also been reported. A non-human primate study suggested that blastomeres may differ substantially in heteroplasmy and that developmental processes can shift heteroplasmy trajectories (Lee *et al.,*

2012). In line with this, a subsequent commentary highlighted that post-implantation segregation could, in some settings, limit the long-term predictive value of a single preimplantation biopsy and contribute to uncertainty around "safe" thresholds (Mitalipov e*t al.,* 2014). Although our data do not indicate such behaviour in human embryos, these observations are relevant when considering scenarios that could increase spatial heterogeneity of mtDNA within the oocyte or early embryo. For example, artificially generated heteroplasmy (e.g., ooplasmic transfer) could, in principle, result in incomplete mitochondrial mixing and uneven representation across blastomeres.

A related concern applies to MRT, where low-level carryover of maternal mitochondria can occur during spindle or pronuclear transfer (Hyslop, 2018). As these procedures involve extensive micromanipulation and transient cytoskeletal perturbation (including actin inhibition with cytochalasin B), and because actin-dependent processes contribute to organelle positioning in oocytes, uneven mitochondrial redistribution is a biologically plausible, yet unproven, mechanism by which spatial heterogeneity could be amplified (Li *et al.,* 2022). To date, systematic assessments of within-embryo heteroplasmy distribution after MRT at the cleavage or blastocyst stage remain scarce, and direct clinical validation is further constrained by regulation: in the United Kingdom, for example, heteroplasmy testing in embryos obtained after pronuclear transfer is not permitted in a clinical context (Hyslop *et al.,* 2025). Consequently, the extent to which post-MRT mtDNA heteroplasmy may vary across embryonic lineages and over development remains an open question, and TE-based sampling could, in principle, be more vulnerable to bias in this setting.

Another proposed limitation of TE biopsy is the hypothesis that mtDNA replication may be initiated earlier or more actively in TE cells than in ICM, potentially altering heteroplasmy in TE relative to the rest of the embryo (Smeets *et al.,* 2023). However, much of the mechanistic basis for this argument is extrapolated from animal studies, including study in mice (Trimarchi *et al.,* 2000) and pigs (Spikings *et al.,* 2007). Thus, there is currently no evidence that mtDNA replication is triggered in certain parts of the human embryo in a way that would change the ratio of mutant and normal mtDNA.

In conclusion, our paired measurements support the feasibility of blastocyst-stage PGT-mt for m.14487T>C and its integration with routine PGT-A on day 5-6 embryos. The observed concordance between TE biopsy and embryo-derived fractions strengthens confidence in TE-based heteroplasmy quantification for embryo selection, while acknowledging that heteroplasmy shifts cannot be excluded in individual cases. When a heteroplasmic embryo is transferred, prenatal confirmation in fetal-derived material (e.g., amniotic fluid) should be considered, and systematic postnatal follow-up remains important to further validate clinical outcomes across larger cohorts.

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**FIGURES, TABLES AND SUPPLEMENTAL DATA**

**Table 1 -** Genetic characteristics of embryos.

**Figure 1 -** Pedigree of the affected family.

**Figure 2 -** Primary concordance analyses with explicit pairings.

**Figure 3 -** Shared-control DABEST-style estimation plot (control = PGT biopsy).

**Figure 4 -** Within-embryo heteroplasmy profiles.

**Figure 5 -** Pairwise Lin's CCC matrix across tissue fractions.

**Supplementary Table 1** - Primer sequences for m.14487T>C to couple PGT-A and PGT-mt

**Supplementary Figure 1 -** PGT-A plots

**Supplementary Figure 2** Embryo parts for tubing

**Supplementary Videos**

**Supplementary Code**

**Table 1.** Genetic characteristics of embryos.

| Embryo number | Chromosomal status | Molecular karyotype | m.14487T>C heteroplasmy, TE biopsy, % | m.14487T>C heteroplasmy | | m.14487T>C heteroplasmy, residual extra cells and cellular fragments recovered from beneath ZP |
|---|---|---|---|---|---|---|
| | | | | TE, % | ICM, % | |
| 1 | (1-22)x2 | XX | 11.4 | 12.9 | 12.2 | 26.4 |
| 2 | (1-22)x2 | XY | 99.6 | 100.0 | 99.6 | 99.6 |
| 3 | (X)x1,(Y)x0 | X0 | 12.0 | 13.5 | | 16.9 |
| 4 | (6)x3,(X,Y)x1 | XY | 92.3 | 90.0 | | 91.4 |
| 5 | (1-22)x2 | XX | 73.2 | 78.7 | 72.3 | 74.5 |
| 6 | (2)x0.5, (11)x0.5, (X)x0.5 | XX | 13.5 | 11.9 | 11.4 | 17.9 |

**FIGURE AND TABLES LEGEND**

**Figure 1.** Pedigree of the affected family. M, muscle; L, lymphocytes; U, urine; n, normal (mutation not detected); D, donor.

**Figure 2. Primary concordance analyses with explicit pairings.** Top row: PGT biopsy vs ICM (scatter and Bland–Altman). Bottom row: PGT biopsy vs remaining TE (scatter and Bland–Altman). Dashed identity lines are shown in scatter panels; Bland–Altman panels show mean bias and 95% limits of agreement (mean ± 1.96 SD).

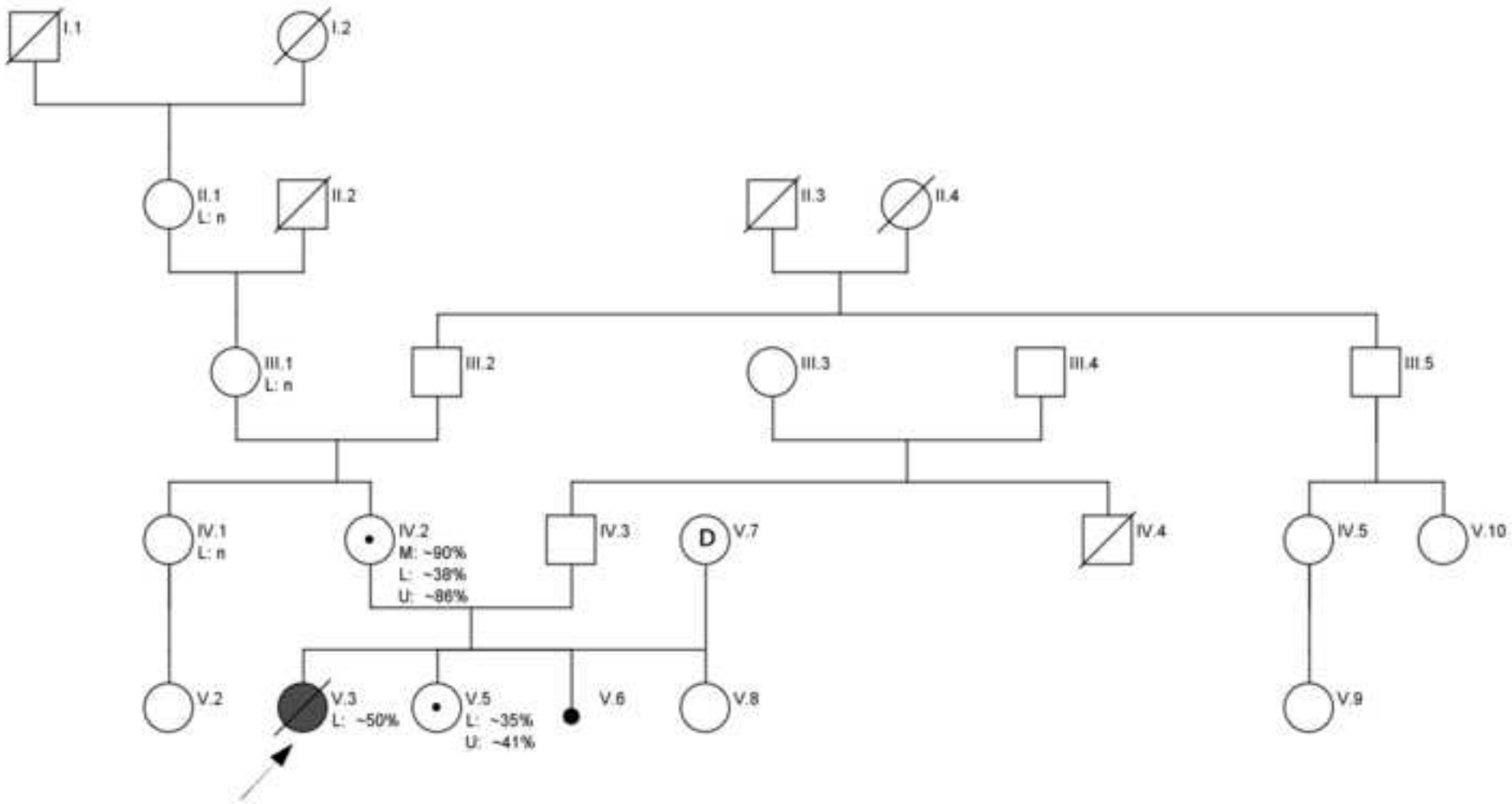
**Figure 3. Shared-control DABEST-style estimation plot (control = PGT biopsy).** Top panel: raw heteroplasmy values by fraction with group means (translucent bars) and jittered embryo-level points. Bottom panel: explicit shared-control reference column at 0, plus bootstrap distributions (10,000 resamples) of mean differences for each target fraction versus PGT biopsy (target − control), with BCa 95% confidence intervals and point estimates aligned under their top-panel groups.

**Figure 4. Within-embryo heteroplasmy profiles.** (A) Slope chart showing het_pct across tissue fractions for each embryo. Near-horizontal lines indicate consistent heteroplasmy within embryos. (B) Deviation of each fraction from the embryo mean. Black diamonds show group means. The EC fraction is shown as an exploratory compartment because it can contain mixed cell debris and excluded fragments.

**Figure 5. Pairwise Lin's CCC matrix across tissue fractions.** Cells show Lin's concordance correlation coefficient for each pair of fractions, computed on embryos where both fractions are available. Values near 1.0 (green) indicate excellent agreement; values below 0.9 (orange/red) indicate potential discordance. N indicates the number of embryos contributing to each comparison.

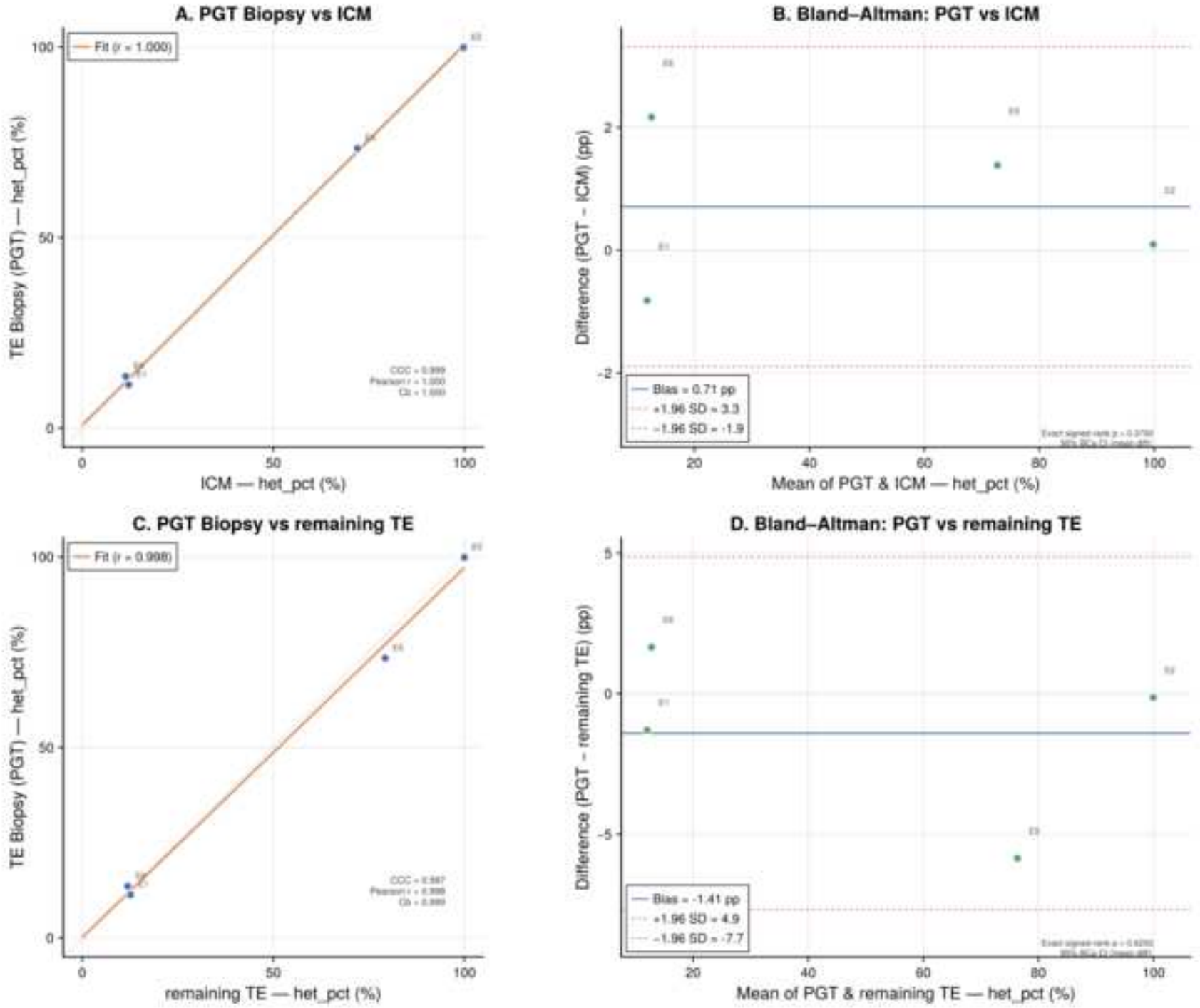
Figure(s)

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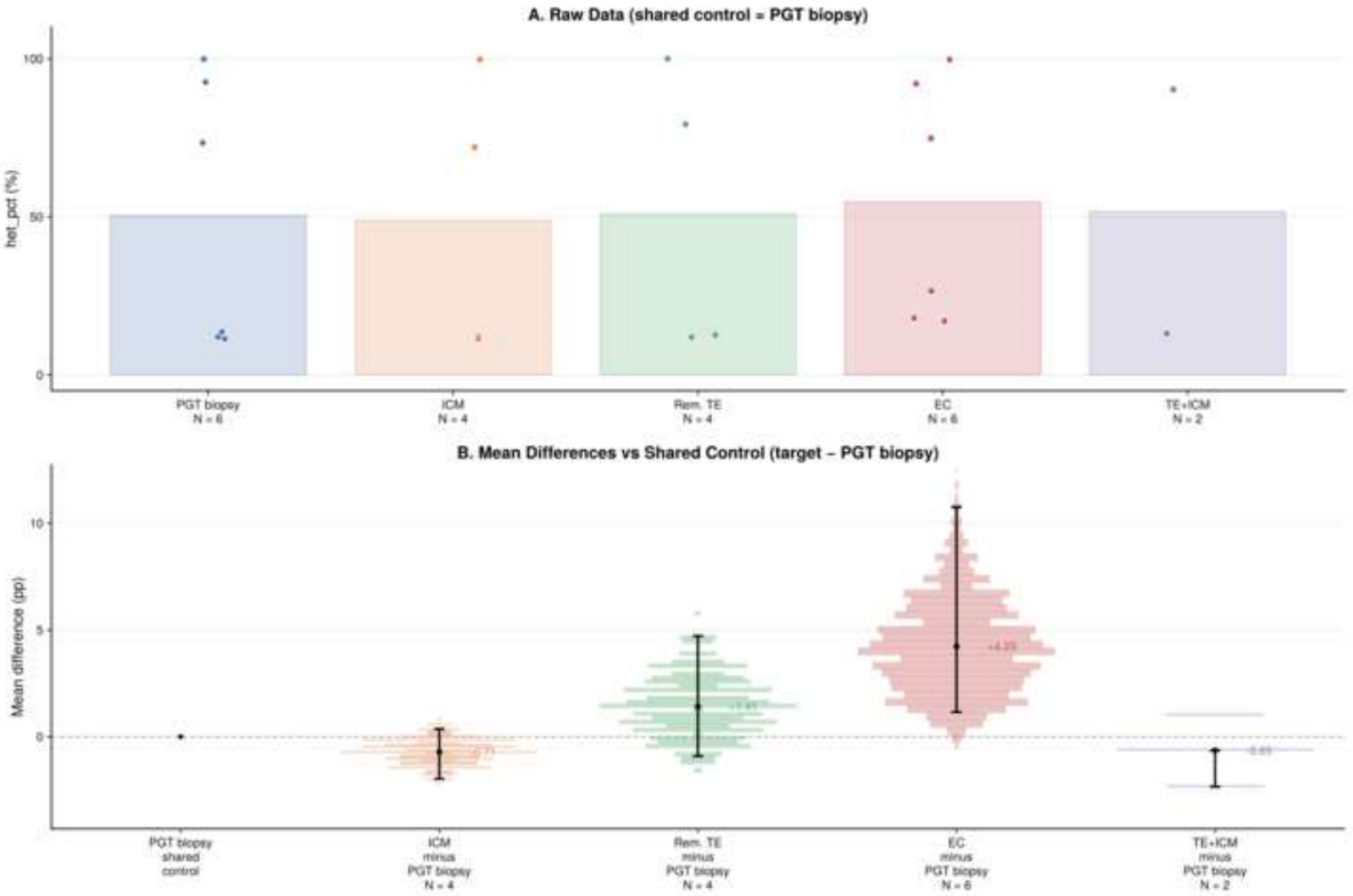
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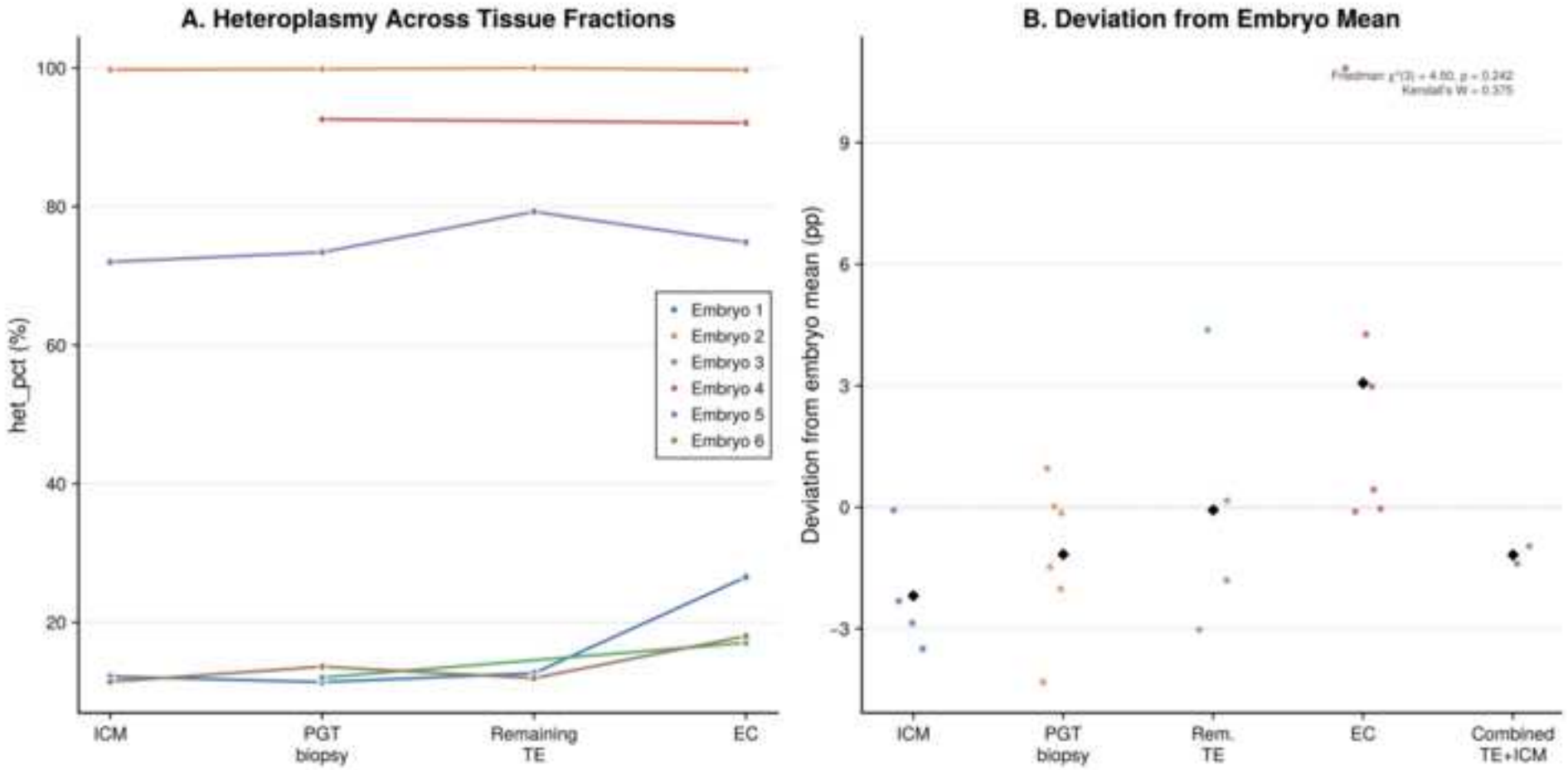
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